MIL-STD-877
NOTICE 1
8 November 1968

MILITARY STANDARD

ANTENNA SUBSYSTEMS, AIRBORNE,

CRITERIA FOR DESIGN AND LOCATION OF

TO ALL HOLDERS OF MIL-STD-877:

1. THE FOLLOWING PAGE OF MIL-STD-877 HAS BEEN REVISED AND SUPERSEDES THE PAGE LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
3 4	9 July 1968	Reprinted without change 4	9 July 1968

2. RETAIN THIS NOTICE AND INSERT BEFORE THE TABLE OF CONTENTS.

FSC 5985

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4.2.1.1 Energy distribution. The distribution of r-f energy (transmitted or to be received) shall be treated as the most important electrical property affecting realization of the overall design objectives. The effects of side lobes, beam widths, nulls, asymmetry of radiation, lobe centering, cross-polarization ratio, directional gain, axial ratio, energy distribution, and any other relevant factors shall be considered with due regard to the type of avionics system involved and its requirements and the particular flight vehicle and its requirements. The ability to provide maximum or adequate coverage establishes the fundamental distribution requirement. Any different distribution necessary for the operation of an avionics system is labeled a special or peculiar distribution requirement.

4.2.1.2 Special energy distribution. Antenna subsystems requiring special energy distribution, such as radar antenna subsystem, navigational, countermeasure, reconnaissance, and communication antenna subsystems shall be designed and located in accordance with applicable detail specifications at the best available locations.

4.2.1.3 Voltage standing wave ratio (VSWR). The voltage standing wave ratio of airborne antenna subsystems, as measured at or corrected to the antenna feed point of the associated avionics system, shall be no greater than that specified in the detailed antenna subsystem specification or avionics system specification. VSWR shall be derived from the measurements performed on, or referred to a transmission line having a nominal characteristic impedance as specified in the detailed antenna subsystem specification or associated avionics system specification.

4.2.1.4 Efficiency. Antenna subsystem design shall provide optimum transfer of energy between the associated avionics systems and free space. Designs which would permit large circulating currents through appreciably resistive conductors, or would permit r-f heating of insulators and dielectrics, shall not be accepted. Each part of the subsystem, including the antenna, cable, waveguide, connectors, insulators, insulating materials, r-f switches, matching devices, interference control devices, radomes, and other pertinent parts shall provide the highest efficiency commensurate with that of the best part development and consistent with other requirements for the part(s) involved. The total subsystem losses, including resistive losses, r-f heating losses, corona losses, and other losses shall be kept as low as practicable, and shall not become an unduly large percentage of the radiated r-f energy.

4.2.1.5 Bonding, conductivity, discontinuities. All metallic elements which are part of or adjacent to the antenna subsystem and not intentionally insulated from each other shall be securely bonded in compliance with MIL-B-5087. The mating surfaces between all metallic parts of the antenna subsystem intentionally designed to be electrically continuous to r-f currents shall be clean metal surfaces free from anodic film, grease, paint, lacquer, or other high resistance film, when necessary to insure negligible r-f impedance between such parts. All portions of the antenna subsystem which are required to carry r-f currents or which would, during operation, have high-surface current densities, shall have a high-surface conductivity. Antenna locations close to windows, skin discontinuities, and other surface irregularities shall be avoided insofar as possible. Requirements of this paragraph shall not necessarily apply to any antenna subsystem having the airframe as an integral part. However, silver-plated rivets may be used in the airframe, when specified by the flight vehicle specification, to improve the surface electrical conductivity of the airframe.

4.2.1.6 Interference control. The design and location of the antenna subsystems shall conform with all the requirements of MIL-E-6051. The subsystems shall be designed and located, including such interference control techniques and devices as necessary, to provide avionics systems performance with negligible interference from subsystems physical and electrical relationships, one to another and each to its surroundings. Design and location of subsystems shall be such that their relationships minimize the need for interference control devices because of the added weight and decreased subsystem efficiency which may otherwise result. The location of subsystems shall be sufficiently removed from external sources and surrounding fixed installations, including other subsystems, jet, fanjet, or propeller driven lift and or life cruise engines; landing gear; etc, to allow the avionics systems to perform without deficiencies from parasitic radiation, shadowing of radiated energy, change in subsystem impedance, propeller modulation, propeller reflection, resonance and antiresonance phenomena, or any other causes. The coupling between subsystems

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shall be kept as low as practicable. Antennas operating in the VHF and UHF frequency ranges when they are similarly polarized and are not used in a multi-purpose installation shall be spaced as far apart as the available space will allow. A minimum spacing of 3/8 wavelength at the lowest frequency of operation of any antenna involved is desirable. A spacing of a full wavelength is sometimes necessary for transmit functions, depending upon the isolation requirements and antenna locations. Subsystem parts, such as fairings and radomes, shall be designed and located in such a way as to minimize undesirable performance effects caused by precipitation static and corona discharge.

4.2.1.7 Radiation pattern coverage. Pattern coverage of an antenna system designed for an avionics system operating at frequencies above 30 megahertz shall be consistent with the particular function of that system. Consideration shall be given to the maneuverability of the flight vehicle on which the antenna is used, possible polarization losses, and possible difficulties involved in antenna location.

4.2.1.8 Multiple-purpose locations. Whenever practical, one location shall be used to accommodate more than one antenna subsystem and serve a maximum number of avionics systems simultaneously. Multiple-purpose locations shall include such filters, coupling devices, matching networks, and interference control devices as necessary to provide satisfactory interference control.

4.2.2 Physical properties. The design and location of the antenna subsystems shall provide optimum compromises between all the subsystems physical properties and the flight vehicle operational characteristics involved. Antenna subsystem physical properties shall include materials, drag, weight, volume and all other relevant properties. Flight vehicle operational characteristics shall include range, altitude, speed, aerodynamic performance, and all other relevant factors. Antenna subsystem electrical requirements such as aperture, special energy distribution, etc, may be sufficient to demand antenna physical properties of a magnitude great enough to contribute need for a highly significant portion of the total size and weight of the flight vehicle. As antenna subsystem size, weight, or drag become greater, the effect on flight vehicle growth shall be considered when determining which design will best meet the overall objective. When the physical properties of a zero drag antenna impose more penalty than those of an external antenna on the operational characteristics of the flight vehicle, the best compromise shall be effected consistent with the overall design objective.

4.2.3 Transmission lines.

4.2.3.1 Coaxial transmission lines. Coaxial cable, coaxial connectors, and accessories for external (lead-in) coaxial transmission lines shall be selected from the preferential lists of MIL-HDBK-216 and shall comply with the requirements of the applicable military specifications.

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